

Deep Learning for Fetal QRS-complex Classification in Noninvasive Fetal Electrocardiogram

Annisa Darmawahyuni
Intelligent System Research Group,
Faculty of Computer Science,
Sriwijaya University
Palembang, South Sumatera
riset.annisadarmawahyuni@gmail.com

Bambang Tutuko
Intelligent System Research Group,
Faculty of Computer Science,
Sriwijaya University
Palembang, South Sumatera
bambangtutuko60@gmail.com

Muhammad Naufal Rachmatullah
Intelligent System Research Group,
Faculty of Computer Science,
Sriwijaya University
Palembang, South Sumatera
naufalrachmatullah@gmail.com

Siti Nurmaini
Intelligent System Research Group,
Faculty of Computer Science,
Sriwijaya University
Palembang, South Sumatera
sitinurmaini@gmail.com

Firdaus Firdaus
Intelligent System Research Group,
Faculty of Computer Science,
Sriwijaya University
Palembang, South Sumatera
virdauz@gmail.com

Muhammad Ardiansyah
Intelligent System Research Group,
Faculty of Computer Science,
Sriwijaya University
Palembang, South Sumatera
ardi412233@gmail.com

Ade Iriani Sapitri
Intelligent System Research Group, Faculty of Computer
Science, Sriwijaya University
Palembang, South Sumatera
adeirianisapitri13@gmail.com

Anggun Islami
Intelligent System Research Group, Faculty of Computer
Science, Sriwijaya University
Palembang, South Sumatera
anggunislami2@gmail.com

Abstract—Fetal QRS-complex generated from an abdominal electrocardiogram (ECG) is an effective noninvasive tool for accurately measuring the fetus well-being during pregnancy. Classification of fetal QRS is significant to avoid fetal heart abnormalities, such as asphyxia, bradycardia, and hypoxia. However, accurately classifying the fetal QRS-complex is hard because the characteristic fetal ECG (fECG) has small amount of signal energy and it has undesired signal components, including recording devices, movement artifacts, and fetal brain activity. The sophisticated fECG signal processing for fetal QRS-complex classification has been used in the literature. However, they mostly use human intervention for feature extraction, leading to time-consume and raising the measurement error. Hence, this study investigates deep learning (DL) that performs automated feature extraction tasks excellently. DL has excellence amount of data scales, which is its ability to automatically learn and improve from data without the need for explicit programming. A sequence-to-sequence learning task was generated using a hybrid four convolution layers as a part of convolutional neural networks and bidirectional long short-term memory (ConvBiLSTM). As the results, the performance is outstanding; the accuracy, sensitivity, specificity, precision, and F1-score were 100%. The observed results show the proposed ConvBiLSTM model is worth evaluating their effectiveness for fetal QRS-complex classification.

Keywords—convBiLSTM, deep learning, fetal electrocardiogram, qrs-complex

I. INTRODUCTION

Fetal electrocardiogram (fECG) analysis from maternal abdominal recordings is a main alternative to the commonly used Doppler ultrasound procedure for fetal heart abnormalities detection [1][2]. The fECG has been used to monitor fetal heart condition from the record of the electrical activity of fetal heart over some time [3]. The fECG

monitoring can be measured by placing the electrodes on the abdomen of expectant mother [3]. Its procedure does not involve the associated risks; therefore, it is more practical than the internal fECG signal.

However, analyzing fECG from the maternal abdominal recordings is a long-time challenge [4]. The main factors affected by maternal ECG (mECG), which it is contaminated by noise, such as power line, electromyogram (EMG) and drifts [2]. Another challenge for the analysis of fECG is the classification of fetal QRS-complex. Fetal QRS-complex is one of the important waveforms for monitoring the fetal heart rate (FHR). The analysis of FHR can be used to diagnose fetal hypoxia that causes congenital disabilities and an indicator to inform fetal distress [5]. The fECG has a fundamental frequency higher than mECG, but the amplitude is much weaker than mECG. Therefore, the heart rate of fetal is faster than the mother. Due to these reasons, the fetal QRS-complexes classification is arduous and still becomes a problem in biomedical engineering.

Computerized methods have been explored for the challenge of fetal QRS classification, i.e., principal component analysis [3], adaptive neuro-fuzzy inference system [6], wavelet transform [7][8], adaptive filtering [9], independent component analysis [10], and so on. Though the abovementioned methods obtained promising results, they extracted features heuristically and still require human intervention (handcrafted).

To overcome the drawbacks, deep learning (DL) arises to automate feature extraction and remove the human expert's dependency [11]. DL performs better, faster, and easier for analyzing and interpreting large amounts of data, it's the best way to automate predictive analytics. DL has varying outstanding algorithms, i.e., convolutional neural networks

(CNN), long short-term memory (LSTM), autoencoder, recurrent neural networks, restricted Boltzmann machines, deep belief networks, and generative adversarial networks. The abovementioned algorithms have been implemented and explored for biomedical engineering tasks, specifically in biomedical signal processing [12][13][14].

Among the DL algorithms mentioned in this investigation, we propose a hybrid of CNN and bidirectional LSTM (BiLSTM). A convolution layer as a part of CNN was used as feature extraction, and BiLSTM as fetal QRS-complex classifier. DL algorithms are powerful in ECG classification tasks [15][16]. Lee *et.al.* [2] have proposed CNN for fetal QRS-complex classification. They proposed seven convolution layers and achieved a sensitivity of 89.06%. Zhong *et al.* [17] have also explored CNN for fetal QRS-complex classification. The differences are the total of used convolution layers. They have used three convolution blocks. The accuracy was achieved by 77.38%. Both previous studies show that using CNN for fetal QRS-complex classification can be considered.

In this paper, we generated the ConvBiLSTM, which consisted of two main parts; convolution layers and BiLSTM. We have first proposed the ConvBiLSTM for adult ECG processing [16]. We generalized the ConvBiLSTM model to fECG processing. The study aims to increase the performance of fetal QRS-complex classification. Compared to previous studies [2][17], the performance results are poorly performed. Hence, it is imperative to investigate the ConvBiLSTM model improvement to increase fetal QRS-complex classification results.

II. MATERIAL AND METHOD

A. Data Preparation

Noninvasive fECG: The PhysioNet/Computing in Cardiology Challenge 2013 has been widely explored to generate the fetal QRS-complexes classification algorithm [1]. The dataset consisted of three data collection sets of one-minute fECG recordings (sets A, B, and C). However, in this study, only set A was used. The total is 63 fECG records which are digitized at 1000 samples per signal per second. The sample records can be presented in Fig. 1. Fig. 1 shows the fECG records. In this investigation, we have only used a single lead for fetal QRS classification among four-lead fECG.

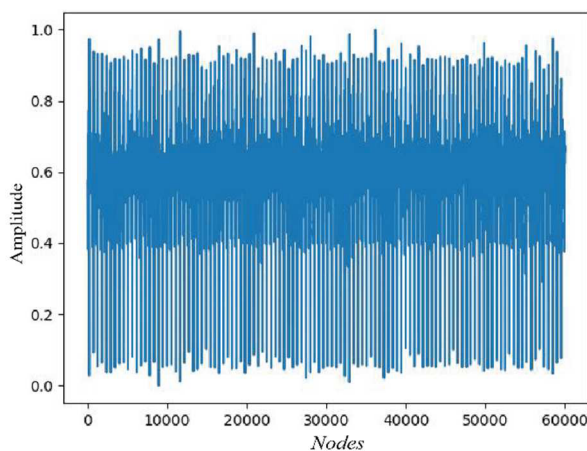


Fig. 1. The sample of fECG signal in a single lead

B. fECG Preprocessing

We have generalized the ConvBiLSTM architecture that was published in our previous works [16][18], to fECG processing. The proposed ConvBiLSTM architecture was robust to adult ECG processing, specifically for ECG delineation tasks. This study used the ConvBiLSTM architecture to detect fetal QRS-complexes. The fECG preprocessing of this paper consisted of: (i) the denoising process to remove the fECG noises using discrete wavelet transform (DWT), and (ii) the beat-to-beat segmentation from the first P-onset to the next P-onset. The details of fECG preprocessing consist of the following two main steps:

- Discrete wavelet transform (DWT) has been implemented to remove fECG recordings from noise or commotion. DWT has powerful and excellences in denoising the signal [19][20]. The dominant frequency is often used to calculate the wavelet disintegration levels. It uses high and low pass elimination to reduce fECG recordings down to the frequency range needed to construct progressive coils [21]. In this investigation, we have explored wavelet families, i.e., *symlets*, *daubechies*, *haar*, and *bior*, to determine which wavelet type would produce the best results for signal denoising. Among the greatest signal-to-noise ratio (SNR) outcomes, *daubechies* wavelet was chosen for ECG signal denoising with SNR of 3.27882 decibels.
- In our experiment, we have segmented beat-to-beat from R-peak. We have segmented the 0.25 seconds (s) to the left from R-peak and 0.45s to the right from R-peak to obtain one beat [22]. The last preprocessing of fECG processing is QRS segmentation. It is the important step before classification using ConvBiLSTM algorithm. This step detects the R-peak from annotation (ground truth). From R-peak, we segment the QRS complex as 0.10 due to its normal width being around 70-100 milliseconds (ms). We have segmented left to R-peak as 0.5s and right as 0.5s. In this step, the labeling process is also done, consisting of QRS-complex and non-QRS-complex.

C. ConvBiLSTM Architecture

The ConvBiLSTM architecture has been generated by using two main parts: the convolution layers of CNN as feature extraction and BiLSTM as a classifier. The convolution layers of CNN have four layers (8, 16, 32, and 64 filters) and adopt the rectified linear unit (ReLU), which has a size of 3 with stride 1. The ConvBiLSTM architecture was trained over 300 epochs, with a learning rate of 10^{-5} and binary cross entropy as the loss metric. We explored Python codes in the Spyder 4.1.5 DL framework and libraries. The resume of ConvBiLSTM architecture can be summarized in Table I.

TABLE I. CONVBiLSTM ARCHITECTURE

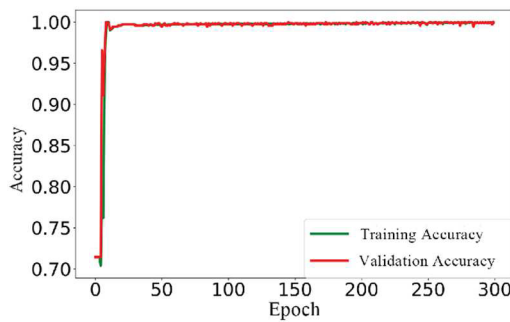
Layers	Learning Rate	Batch Size	Epochs
Convolution 8 x 3, strides= 1 + ReLU	10^{-5}	2	300
Convolution 16 x 3, strides= 1 + ReLU			
Convolution 32 x 3, strides= 1 + ReLU			
Convolution 64 x 3, strides= 1 + ReLU			
BiLSTM			

III. RESULTS AND DISCUSSION

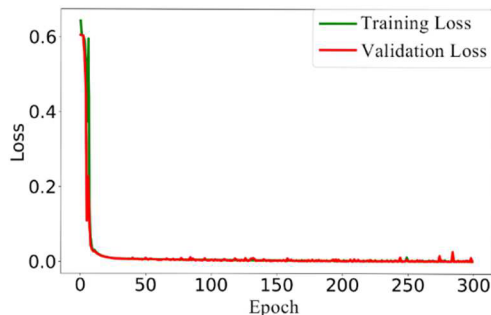
A total of 63 fECG records have been used to generate the fetal QRS-complexes classification model. It consisted of 50, and 13 for training and validation set, respectively. To reduce the data leakage problems, we have divided the fECG records based on patient (record-based). In this study, we have 317,000 fetal QRS and 126,800 fetal non-QRS. The results of each class can be listed in Table 2. Table 2 shows all the metrics were achieved 100%.

TABLE II. THE PERFORMANCE RESULTS OF FETAL QRS-COMPLEX CLASSIFICATION BASED ON TESTING SET

Metrics	Performance Results (%)		Average (%)
	<i>Fetal QRS</i>	<i>Fetal Non-QRS</i>	
Accuracy	100	100	100
Sensitivity	100	100	100
Specificity	100	100	100
Precision	100	100	100
F1-score	100	100	100



a. Accuracy curve



b. Loss curve

Fig. 2. The learning curves of the ConvBiLSTM model

To present the performance of the training and validation set over the experience of accuracy and loss metrics related to learning of ConvBiLSTM model, we present the learning curve (accuracy and loss curves) (Fig. 2). According to these curves, we measure time or progress (epoch) in the x-axis and measure accuracy/loss performance in the y-axis. The higher accuracy and the lower loss, the ConvBiLSTM model performance shows better. The performance of model grows over time, and the model improves with experiences or

learning of features. The learning curves can evaluate the issues in the model behavior, such as underfitting (high bias) and overfitting (high variance). From the learning curves presentation, we can avoid those problems and find the right bias (good fitting).

This study has also used a confusion matrix (CM) to visualize and summarize the performance of a classification algorithm. It was presented with two dimensions table, which consisted of actual and predicted and a set of classes in both dimensions. The visualization of CM for binary classification (fetal QRS-complex as 0, and non-fetal QRS-complex as 1) in this task can be presented in Fig. 3.

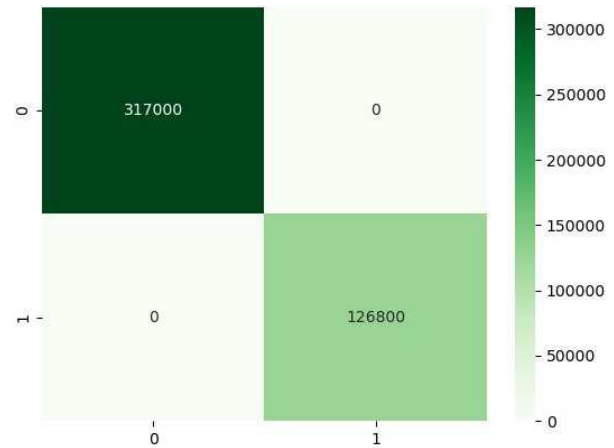


Fig. 3. The CM of ConvBiLSTM model

As a result, we have compared the expert annotation (ground truth) and ConvBiLSTM-predicted of fetal QRS. The fetal QRS and non-fetal QRS were colored red and blue, respectively. As shown in Fig. 4, the differences between expert annotation and ConvBiLSTM are no different. For the sample beat visualized in Fig. 4, ConvBiLSTM predicted model successfully classified the fetal QRS and non-fetal QRS.

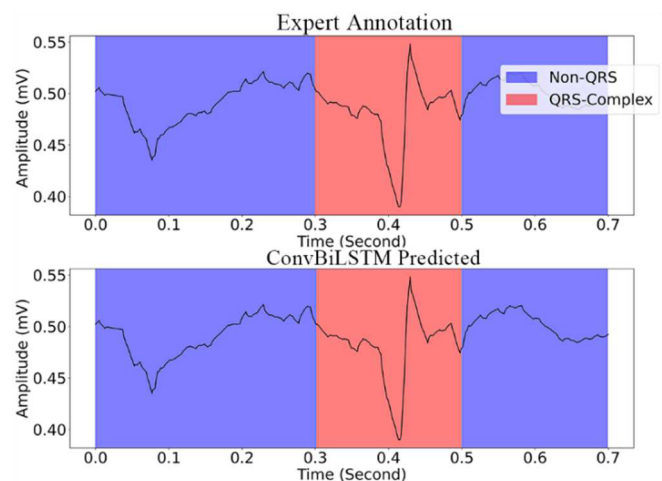


Fig. 4. The comparison of expert annotation and ConvBiLSTM predicted for fetal QRS-complexes classification

In this study, we have compared our proposed model to other previous studies that have explored fetal QRS-complex classification task DL-based (Table 3). With the same

experimented dataset (i.e., Noninvasive fECG: The PhysioNet/Computing in Cardiology Challenge 2013), our proposed model outperformed the previous research results with 100% accuracy, sensitivity and precision.

TABLE III. THE COMPARISON RESULTS WITH PREVIOUS RESEARCH FOR QRS-COMPLEX CLASSIFICATION BASED ON DL

Authors	Method	Performance (%)		
		Accuracy	Sensitivity	Precision
Lee et al., [2]	CNN	-	89.06	-
Zhong et al., [17]	CNN	-	80.54	75.33
Krupa et al., [23]	MobileNet	91.32	89.52	94.37
Proposed Model	CNN-BiLSTM	100	100	100

IV. CONCLUSION

This study has generalized the ConvBiLSTM model for fetal QRS-complex classification. The model was generated using four convolution layers (8, 16, 32, and 64 filters) as feature extraction and BiLSTM as a classifier. Both DL algorithms have excellences to employ for the sequence-to-sequence learning task. With the powerful ConvBiLSTM model, we have achieved 100% accuracy, sensitivity, specificity, precision, and F1-score for the fetal QRS-complex and non-fetal QRS-complex from a total of 63 records in a single-lead experiment. The hybrid of the convolution layer and recurrent network algorithm provides new insight into assessing the signal quality of fECG. In addition, in the future, the performance results are worth evaluating their effectiveness for fetal QRS-complex classification.

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